K. SIDDHARTH UDUPA

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SUMMARY

Second-year BCA student with hands-on experience building and deploying production-style full-stack web applications using TypeScript, Next.js, React, Node.js, Express.js, PostgreSQL, and Drizzle ORM. Experienced in authentication, AI-powered RAG systems, tool orchestration, secure file uploads, vector search, Docker-based deployments, and backend architecture. Seeking a Software Engineering Internship focused on backend or full-stack development.

TECHNICAL SKILLS

Languages	TypeScript, JavaScript (ES6+), SQL, HTML5, CSS3
Frontend	React, Next.js (App Router), Tailwind CSS, Zustand
Backend	Node.js, Express.js, Hono, grammY (Telegram Framework), REST APIs
Databases	PostgreSQL, Drizzle ORM, pgvector
Auth & Security	Better Auth, JWT, Google OAuth 2.0, bcrypt, Helmet, Rate Limiting
AI	Google Gemini, RAG, Tool Orchestration, Vector Embeddings, Tavily Search
DevOps & Tools	Git, GitHub, Docker, Linux, Vercel, Cloudflare, node-cron

PROJECTS

Atlas AI – Intelligent Telegram Assistant

Tech: TypeScript, Express.js, grammY, Google Gemini, PostgreSQL, Drizzle ORM, Tavily Search, Google OAuth

- Built an AI-powered Telegram assistant providing on-demand market intelligence briefings, email summaries, and conversational onboarding.
- Integrated Google Workspace (Calendar & Gmail APIs) via OAuth to manage user events and fetch unread emails within Telegram.
- Engineered dynamic personalization using entity extraction and a daily decay job (`node-cron`) to maintain evolving user interest scores.
- Architected a backend tool orchestrator in Express.js to route Gemini AI function calls to external APIs and PostgreSQL.

Axiom – AI RAG Web Application

Tech: Next.js 16, React 19, PostgreSQL, Drizzle ORM, Better Auth, Gemini, pgvector, Docker

- Built a multi-user AI assistant with streaming responses. Live: axiom-rag-app.vercel.app | GitHub: github.com/siddharth-udupa/Axiom
- Implemented Retrieval-Augmented Generation using pgvector embeddings.
- Designed a planner executor summarizer pipeline for web-assisted responses.
- Implemented automatic model fallback for API rate limits and containerized with Docker for Vercel deployment.

Strava Clone *(In Progress)*

Tech: Next.js, React, TypeScript, PostgreSQL, Drizzle ORM, Leaflet GitHub: github.com/siddharth-udupa/Strava-clone-app

- Built a Strava-inspired fitness platform with a production-oriented backend architecture.
- Implemented secure authentication and infinite-scrolling activity feed using cursor-based pagination.
- Added GPX uploads, route rendering, interactive maps, and activity previews.
- Designed secure GPX validation before parsing to reduce malformed input risks.

Attendance Calculator

Tech: Node.js, Express.js, PostgreSQL, Drizzle ORM, Tailwind CSS

- Built and deployed a full-stack attendance tracking application with optimistic UI updates and local caching. Live: ad-cal.vercel.app | GitHub: github.com/siddharth-udupa/Attendance-calculator
- Secured backend using JWT authentication, bcrypt, Helmet, rate limiting, and input validation with Drizzle ORM schema.

OPEN SOURCE EXPERIENCE

The Untested API

- Wrote unit/integration tests using Jest & Supertest; identified and fixed bugs through test-driven debugging.

Engineering Assignment

- Implemented and tested a REST endpoint with validation following the project's existing architecture.

EDUCATION

Bachelor of Computer Applications (BCA) Expected Graduation: 2028

INTERESTS

Backend Engineering • Full-Stack Development • Artificial Intelligence • System Design • Open Source